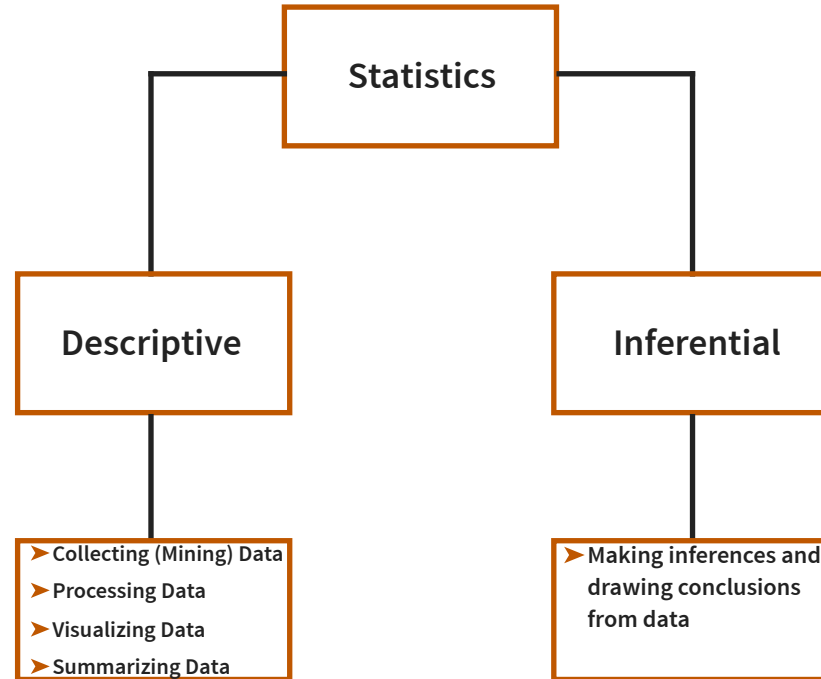


Confidence Intervals

SDS 320E

Inferential Statistics



- **Inferential Statistics:** *the process of making judgments about a population based on properties of a sample randomly drawn from the population*

Basics of Intervals

Parameter Statistic Refresher

! Goal of Statistics

Generally, the goal of statistics is to determine some characteristic about the population of interest.

- Generally, the characteristic of the population that we are interested in is a *parameter* parameter is only know when we take a census

i Reminder

We use a *statistic* to estimate a *parameter*

Parameter Statistic Refresher Cont.

Example

Suppose the FDA requires more than 50% of possible patients to see improvement for a proposed drug to be approved. A pharmaceutical company in Austin, Tx has developed a new Alzheimer's drug that is supposed to improve mental cognition, and they want to get the drug approved by the FDA.

What is the population of interest?

What is the parameter of interest?

How would we estimate the parameter of interest?

Suppose _____, should the drug be approved by the FDA?

Estimators

➤ **Estimator:** a statistic that estimates a parameter

➤ Ex: $\bar{X}$ is an estimator for μ

➤ **Point Estimate:** the numerical value of an estimator once a sample has been taken

➤ Ex: Suppose a sample is taken and yields the observations 1, 2, and 3. Then,
 $\bar{x} = \frac{1+2+3}{3} = 2$ is the *point estimate*.

! Issue

We simply can not use a point estimate from one sample and generalize it to the entire population.

An Aside: Properties of Estimators



A Better Way of Estimating Parameters

Example

Suppose you are interested in renting an apartment in West Campus and someone tells you they pay \$1,500 for a one-bedroom apartment. Is \$1,500 a good estimate for what ALL one-bedroom apartments in West Campus cost?

What would be a more informative way to estimate the typical price of a one bedroom apartment in West Campus?

Idea

We need to account for the variability in taking a sample when estimating a parameter

A Better Way of Estimating Parameters Cont.

Example

Suppose the FDA requires more than 50% of possible patients to see improvement for a proposed drug to be approved. A pharmaceutical company in Austin, Tx has developed a new Alzheimer's drug that is supposed to improve mental cognition, and they want to get the drug approved by the FDA. The company just finished its clinical trial and found 50.5% of those in the sample had improved mental cognition.

Suppose, based upon the clinical trial results, a statistical interval for the proportion of Alzheimer patients who would see improvement is found to be (47.2%, 53.8%). Should the drug be approved?

Basics of Intervals

Idea of Intervals

Rather than estimating a parameter with one number, it is more informative to estimate it with an interval of reasonable values.

➤ **Formula for Intervals:** The basic formula to construct a statistical interval for a parameter is given by

which yields

or

For short hand, we will use LB and UB for lower and upper bound, respectively

Basics of Intervals: Error and Length

- **Margin of Error (E):** *represents the uncertainty in using an interval to estimate the true value of the parameter*
 - **Length of Interval (L):** *refers to how “wide” the interval is, i.e the distance between the UB and LB*
 - The *Margin of Error* and *Length* have the following relationships
 - **Point Estimate:** *given an interval (LB, UB) , we can determine the *point estimate* value using the following*
-

Basics of Intervals Cont.

Example

Suppose a statistical interval for σ^2 is found to be $(3, 7)$. Find the following.

1. The length of the interval
 2. The margin of error for the interval
 3. The point estimate s^2
-

Confidence Intervals

Confidence Intro.

Example

Suppose you are interested in renting an apartment in West Campus and someone tells you they are “confident” that the average rent for a one-bedroom is between (\$0, \$1 Billion).

Are they correct?

Is their interval helpful?

- We have been referring to “statistical” intervals, but now we want to formally define the intervals
 - The goal is to construct an interval that balances “confidence” and informativeness

Confidence Intervals Intro.

Idea

Instead of using a single point to estimate the parameter, we use a range of plausible values i.e an interval. Along with that range, we report how “confident” we are that the interval captures the true value of the parameter.

- A useful interval should therefore be:
 - “confident” enough to reliably capture the parameter
 - narrow enough to still be informative
- **Confidence Interval:** *an interval estimate of a parameter in which we include how “confident” we are that the interval contains the true value of the parameter we are estimating*

What Confidence Means

Example

Suppose we are constructing 90% “confidence” intervals for the population mean μ . Further suppose that the true value of $\mu = 50$.

Confidence Level

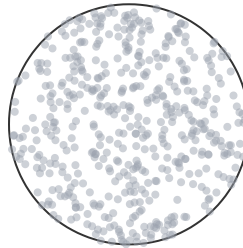
95%

▼

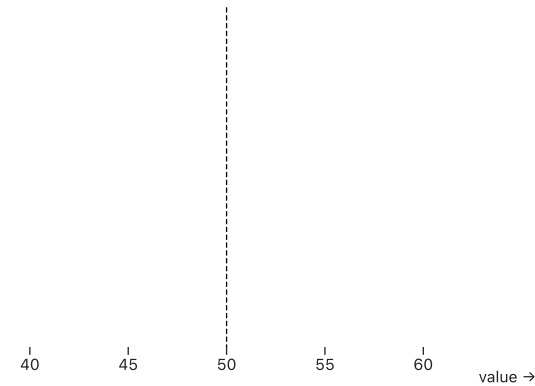
Draw Sample and Construct Interval

Draw 100 Samples

Population



Click "Draw Sample" to construct a 95% confidence interval for $\mu = 50$



Confidence Cont.

- **Confidence:** *will refer to % of times the method used to construct the interval captures the true value of the parameter if we took repeated samples*
- Generally, researchers construct 90%, 95%, or 99% confidence intervals the confidence % is also referred to as the confidence level

Example

Suppose you are interested in renting an apartment in West Campus and someone tells you they are “confident” that the average rent for a one-bedroom is between (\$0, \$1 Billion). What confidence level do we think this interval is?

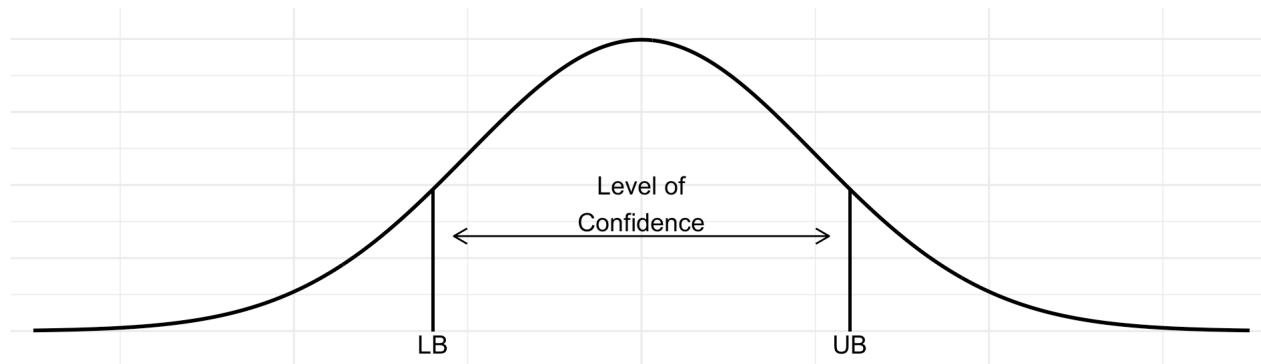
- Thus, confidence level and interval length have the following relationship:

Constructing Confidence Intervals

i Reminder

1. The basic formula to construct a statistical interval for a parameter is given by: *Point Estimate* $\pm$ *Error*
2. Margin of Error represents the uncertainty in using an interval estimate

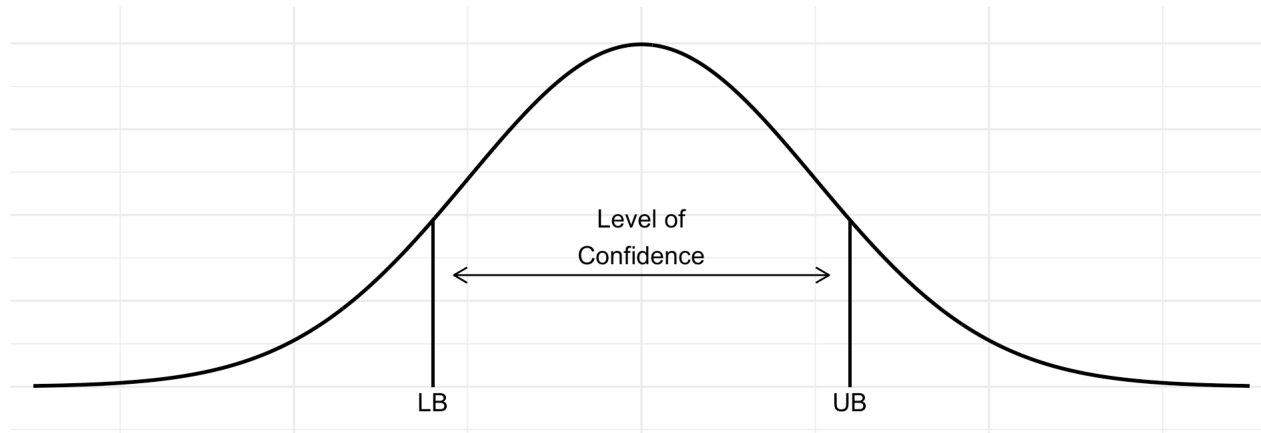
- In general, what statistical tool do we use to quantify uncertainty?
- What describes all possible values of a statistic that can be obtained?



CI for μ with σ known

i Central Limit Theorem (CLT)

Suppose X is distributed with mean μ and standard deviation σ . Then, for a “sufficiently large” sample size n , the sampling distribution of $\bar{X}$ is $\bar{X} \sim \mathcal{N}(\mu, \sigma/\sqrt{n})$.



i Confidence Interval for μ with known σ

Suppose σ is known. Then, a $(1 - \alpha)\%$ confidence interval for μ is given by

CI for μ with σ known Cont.

Example

A hospital wants to estimate the average systolic blood pressure of its adult patients. Suppose it is known that the population standard deviation is 16 mmHg. The hospital took a random sample of 64 patients and found that the mean systolic blood pressure of was 128 mmHg for the sample. The hospital wants to construct a 95% confidence interval for the mean systolic blood pressure for the population.

T-Confidence Interval

Issue with CI for μ with σ known

 Reminder: CI for μ with known σ

Suppose σ is known. Then, a $(1 - \alpha)\%$ confidence interval for μ is given by $\bar{x} \pm Z_{\alpha/2} \cdot \sigma/\sqrt{n}$

- The goal of a confidence interval for μ is to
 - Do we know the value of μ ?

 Issue

Is it likely that we know σ ?

- What can we use to estimate σ ?

CI for μ with unknown σ

- We used the CLT, $\bar{X} \sim N(\mu, \sigma/\sqrt{n})$ to get the confidence interval for μ
 - However, now, we have to estimate σ with s
 - Thus, the sampling distribution used to construct the confidence interval for μ will no longer be normally distributed

t Confidence Interval

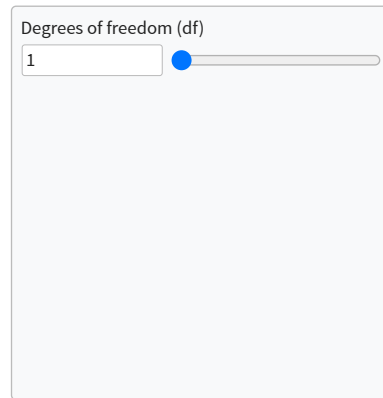
Suppose σ is unknown. Then, a $(1 - \alpha)\%$ confidence interval for μ is given by

where t_α is the α percentile of the $t(\quad)$ distribution. *df* stands for degrees of freedom and parameterizes the t-distribution

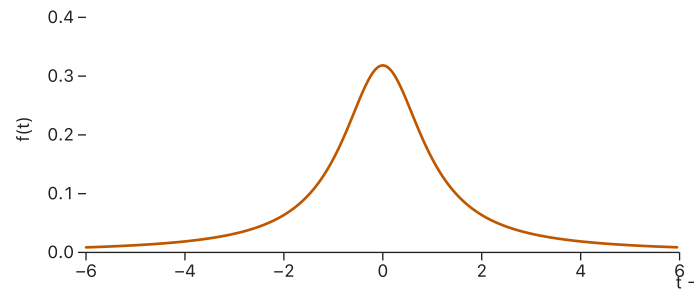
Assumption for the Confidence Interval:

t-Distribution

- The t-distribution is parameterized by the “degrees of freedom” (df)
 - it is very similar to the standard normal distribution but it has “fatter” tails
i.e. more values are in the extreme



t-distribution (df = 1)



- There are percentile tables for the t-distribution: [t-Table link](#)

Assessing Distributional Assumption

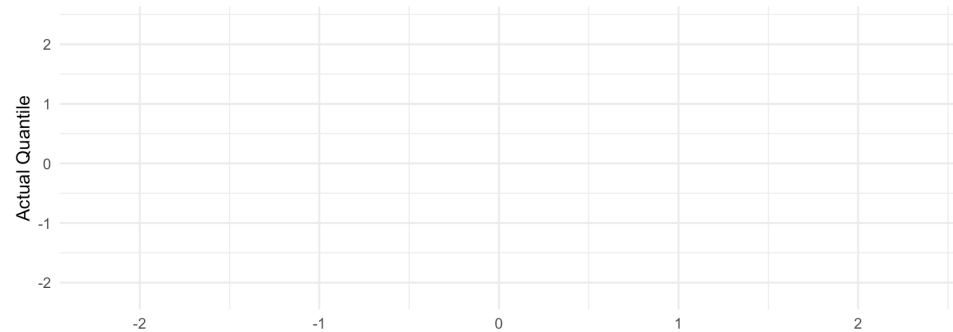
- The assumption for the t confidence interval was that our data comes from a normal distribution
 - One way to visually check whether our data follows a certain distribution is to use a QQ-plot

💡 QQ-Plot

A QQ-plot plots the percentiles/quantiles of our data against the *theoretical quantiles* if our data came from the assumed distribution

If our data actually comes from the assumed distribution, what would be the relationship between the *theoretical quantiles* and the *actual quantiles*?

If our data actually comes from the assumed distribution, what would we expect to see when we plot the *theoretical quantiles* against the *actual quantiles*?



Using a QQ Plot

💡 QQ-Plot interpretation

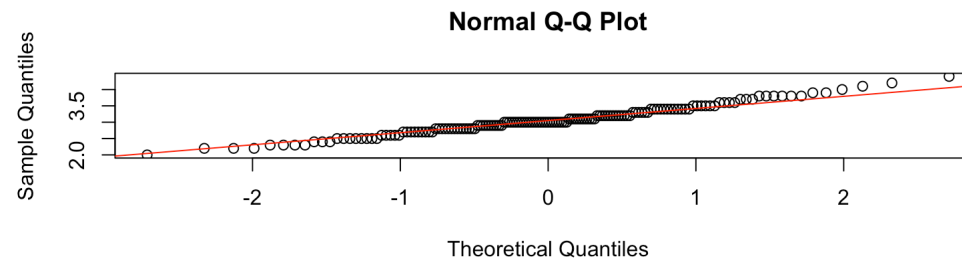
If most of the points for the QQ-plot fall close to the line, then we say it is reasonable to assume our data came from the proposed distribution.

➤ To construct a QQ-plot in R, we can use the `qqnorm()` function or `qqplotr` package

the default QQ-plot is for the normal distribution, but QQ-plots can be for other distributions: uniform, exponential, beta, ect.

➤ Base R ➤ `ggplot`

```
1 qqnorm(iris$Sepal.Width)
2 qqline(iris$Sepal.Width, col = "red")
```

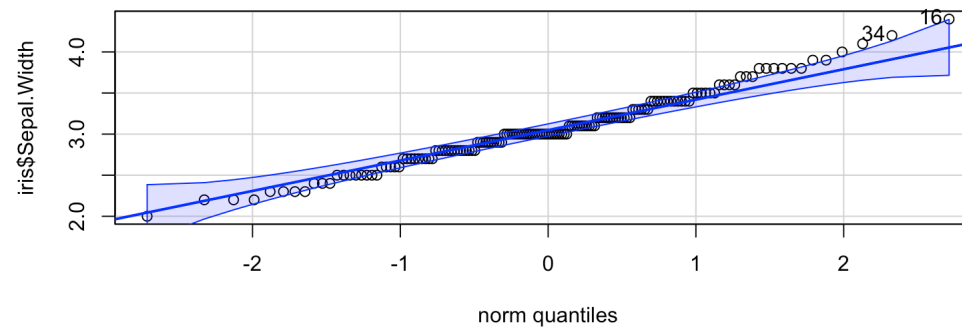


Using a QQ Plot Cont.

- To assess if points are close to the line, confidence bands are often used with QQ-plots
 - For Base plot, use must use the `car` package

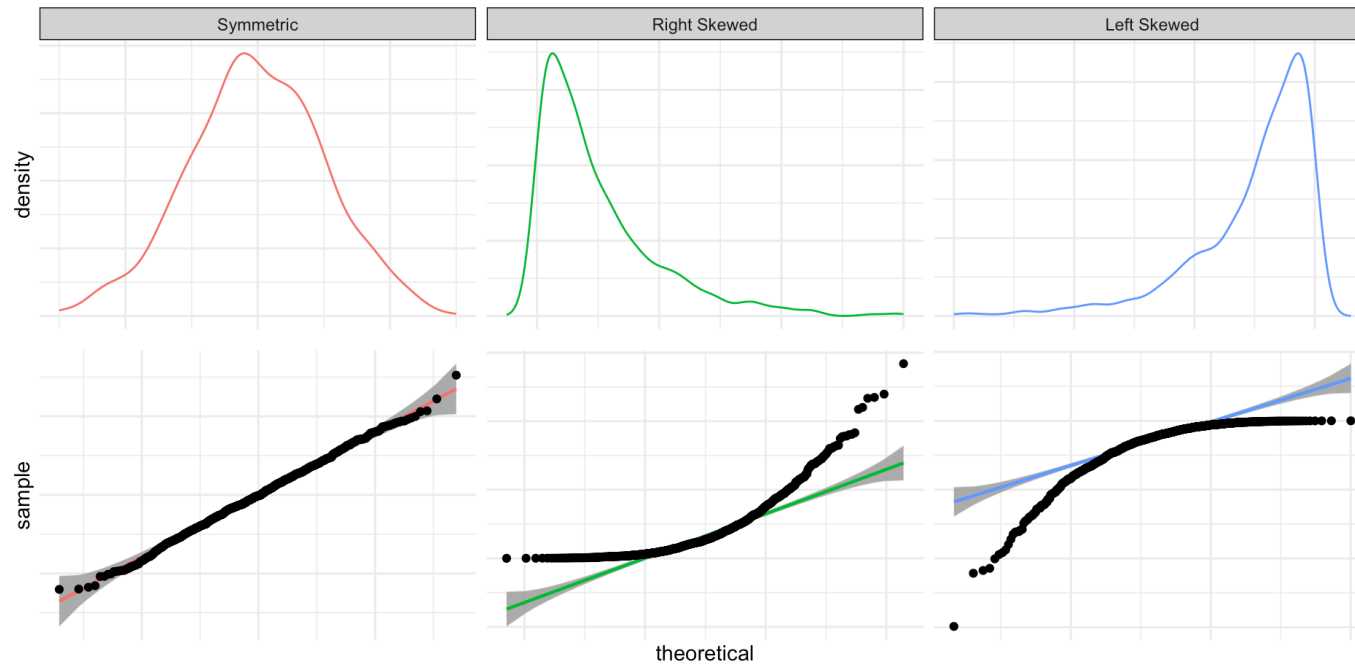
➤ Base R ➤ `ggplot`

```
1 car::qqPlot(iris$Sepal.Width)
```



QQ-Plot for Normal Distribution

- For a normal distribution QQ-plot, we expect it to be sensitive to skewness



Why QQ-plots?

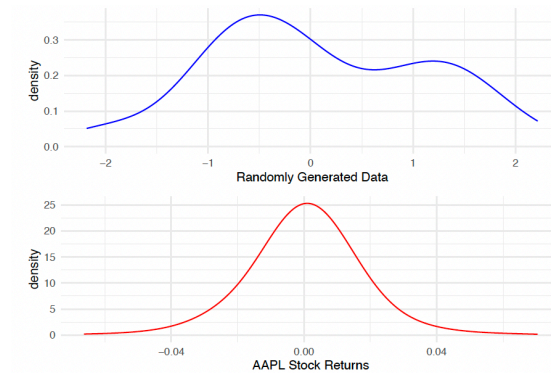
- A QQ-plot gives us a way to check distributions, but the idea under it may seem confusing. What are other ways to check distribution?



- Why not use those instead of a QQ-plot?

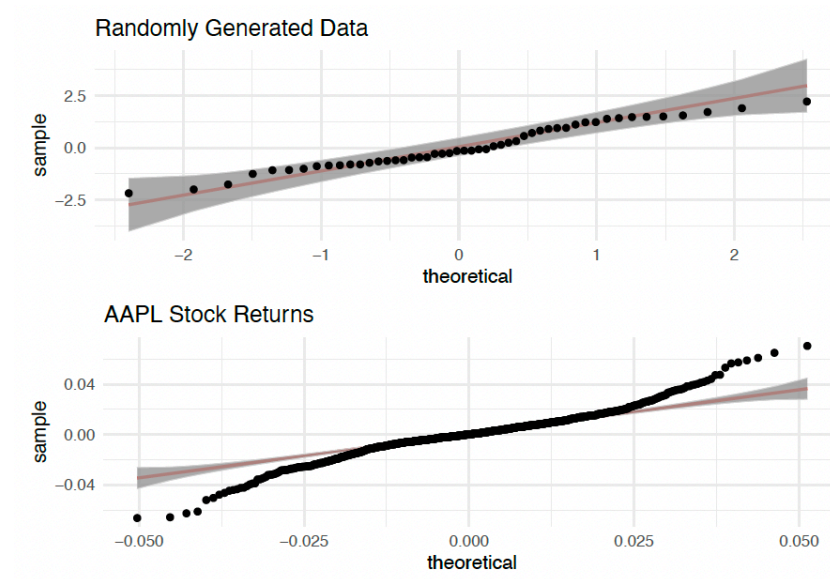
Example

Consider random generated data and also Stock Return data for Apple in 2018. Which data appears to follow the normal distribution?



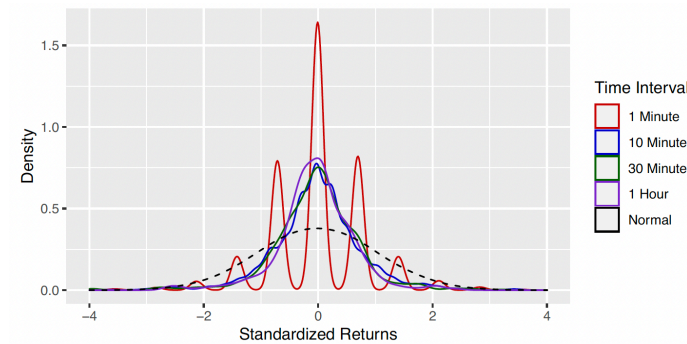
Why QQ-Plots Cont.

- However, if we look at the QQ-plots for each, we find a different story. The Apple returns clearly are not normal.



Why QQ-Plots Cont.

- Why is this the case?
 - Distribution plots focus on the center of the data and can often miss displaying the tails *remember over/under plotting?*
- Stock returns are famous for having excess information in the tails. This becomes worse as we consider intra-day returns.



CI for μ Example

Example

A hospital wants to estimate the average bacterial count remaining on surgical instruments after a sterilization procedure. The hospital took a random sample of 25 sterilized instruments and found that the mean bacterial count was 6.8 CFU, with a sample standard deviation of 4.5 CFU. The hospital wants to construct a 90% confidence interval for the mean bacterial count for all instruments sterilized using this procedure.

```
1 sterilization_data
```

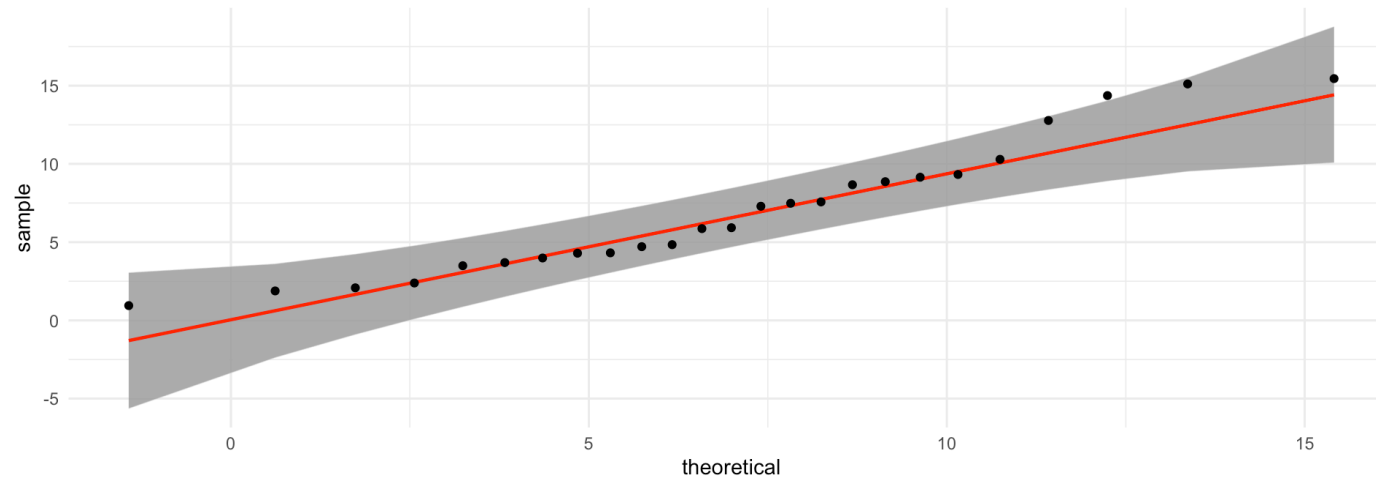
instrument	CFU
1	4.2943775
2	5.8643477
3	14.3672637
4	7.2935651
5	7.5729546
6	15.1104568
7	9.1492481
8	0.9453476
9	3.6936825
10	4.8401088

Before, we calculate the interval, what must verify the assumption for the t-interval. What is the assumption for the t-confidence interval?

How we do check the assumption?

CI for μ Example Cont.

```
1 library("qqplotr")
2 ggplot(aes(sample = CFU), data = sterilization_data)+
3   stat_qq_band(color = "lightgrey")+
4   stat_qq_line(color = "red") +
5   stat_qq_point()
```



Does the assumption seem reasonable?

CI for μ Example Cont.

Example

A hospital wants to estimate the average bacterial count remaining on surgical instruments after a sterilization procedure. The hospital took a random sample of 25 sterilized instruments and found that the mean bacterial count was 6.8 CFU, with a sample standard deviation of 4.5 CFU. The hospital wants to construct a 90% confidence interval for the mean bacterial count for all instruments sterilized using this procedure.

Now that we verified the assumptions for the interval, we can continue to construct the t-confidence interval.

t Confidence Interval in R

- To get a t confidence interval for μ in R, we can use the `t.test()` function. It has the following arguments:
 - `x`: the numeric vector of data
 - `conf.level`: the confidence level of the interval (default is 0.95)
 - `alternative, mu`: used for hypothesis tests, so we can leave these at their defaults for now
- `t.test()` returns a lot of output, so we use `$conf.int` to extract only the interval

```
1 # 90% CI for the mean bacterial count (CFU)
2 t.test(sterilization_data$CFU, conf.level = 0.90)$conf.int

[1] 5.559549 8.422736
attr("conf.level")
[1] 0.9
```

The End

